上海音乐学院
SHANGHAI CONSERVATORY OF MUSIC

认知障碍患儿康复训练答题系统

——基于脑机接口和眼动仪

商业计划书

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Business Plan for the Rehabilitation Training Answering System for Children with Cognitive Disorders Based on Brain - Computer Interface and Eye Tracker

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Part 1: Project Overview

We are committed to developing a rehabilitation training answering system for children with cognitive disorders, which is based on brain - computer interface and eye - tracker technologies. This system utilizes a multi - sensory collaborative training strategy. Through real - time brainwave detection and convenient eye - movement operation, it provides personalized and efficient rehabilitation training solutions for children with cognitive disorders. The aim is to significantly improve the rehabilitation effect and reduce the labor cost during the rehabilitation training process.

Part 2: Market Analysis

(I) Market Size

The number of children with cognitive disorders is large and on the rise. According to relevant statistics, the number of such children in China is increasing continuously, which leads to a growing demand for high - quality rehabilitation training services and products. Currently, traditional rehabilitation training methods have many limitations, and the market for rehabilitation training systems supported by new technologies remains to be fully developed, with an extremely large potential market size.

(II) Target Customers

1. Medical Institutions: Children's hospitals at all levels, rehabilitation hospitals, and the rehabilitation departments of general hospitals. These institutions urgently need advanced rehabilitation training equipment to improve treatment effectiveness and efficiency and reduce labor input.

2. Families of Children with Disorders: Parents of children with cognitive disorders are eager to find effective rehabilitation training methods to help their children improve cognitive functions, enhance self - care ability and social adaptability, while expecting the rehabilitation cost to be controllable.

(III) Market Demand

Traditional rehabilitation training systems have high labor costs and limited rehabilitation effects, failing to meet the increasing rehabilitation needs. With the in - depth research on cognitive disorders, the multi - sensory collaborative training strategy has been proven to effectively improve the rehabilitation effect. Therefore, parents and medical institutions have a strong demand for rehabilitation training systems based on such advanced technologies. In 2022, the National Health Commission issued the "Specifications for Autism Screening and Intervention Services for

Children Aged 0 - 6 (Trial)", clearly requiring local governments to implement the disease prevention and control strategy of early screening, early diagnosis, and early intervention. This policy direction is consistent with our efforts to build a rehabilitation training system for children with cognitive disorders using brain - computer interface and eye - tracker technologies, both aiming to improve children's cognitive rehabilitation services. At the same time, the increasing social attention to children with cognitive disorders has also led to an increased market demand for better rehabilitation solutions.

(IV) Competition Analysis

Currently, in the market, there are relatively few rehabilitation training products for children with cognitive disorders. Most of them are traditional visual - stimulation or simple answering systems. Among the few products involving emerging technologies, there are deficiencies in multi - sensory collaborative training and personalized customization. Our system, with the help of brain - computer interface and eye - tracker technologies and combined with a multi - sensory collaborative training strategy, has significant advantages in terms of personalization, convenience, and low cost, which can effectively distinguish it from competitors.

Part 3: Technical Solution

(I) System Architecture

The entire rehabilitation training answering system consists of a data collection layer, a data processing layer, a training application layer, and a user interaction layer. The data collection layer uses brain - computer interface devices and eye - trackers to collect children's brain - electrical signals and eye - movement data in real - time. The data processing layer analyzes and processes the collected data using advanced algorithms to extract key features. The training application layer provides personalized training tasks and content for children according to the analysis results. The user interaction layer is responsible for achieving friendly interaction between children and the system, including answering operations and feedback display.

(II) Hardware Equipment

1. Brain - Computer Interface Device: We select advanced non - invasive brain - electrical sensors on the market. These sensors can obtain brain activity information only on the scalp surface without penetrating the brain, featuring easy operation and high safety. Notably, its cost is lower than that of similar products. By

analyzing brain - electrical data, information such as children's relaxation and concentration levels can be obtained to trigger question - type changes and provide a basis for personalized training. This low - cost equipment selection not only meets the needs of large - scale promotion but also reduces the use cost for customers, enhancing the product's competitiveness in the market.

2. Eye - Tracker: We adopt common eye - tracking technology. Although its accuracy is not as high as that of high - end products, it can meet the basic requirements of gaze point and eye - movement trajectory capture, which is sufficient for the answering operations of children with cognitive disorders. Compared with high - precision eye - trackers, it can significantly reduce the procurement cost, making the price of the entire rehabilitation training system more affordable.

(III) Software System

1. Training Content Design: Based on the principle of multi - sensory collaboration, a variety of training content is designed. For example, specific sounds (such as animal calls, musical instrument sounds, etc.) are played, and children need to select the matching patterns from multiple visual options (such as corresponding animal pictures, musical instrument pictures, etc.). At the same time, tactile

feedback (such as vibration) can be added to fully activate multiple sensory channels.

2. Generation of Personalized Training Plans: According to children's individual differences (such as cognitive level, age, and disease severity) and real - time collected data, personalized training plans are automatically generated. The system dynamically adjusts the training difficulty, question types, and content to adapt to children's rehabilitation progress.

3. Data Analysis and Feedback System: In - depth analysis is carried out on a large amount of data generated during the training process, including answering accuracy, response time, and attention concentration. The training results are presented to parents and medical staff in a visual way to provide data support for subsequent training adjustments.

Part 4: Core Technologies

(I) Brain - Computer Interface Technology

1. Brain - Electrical Signal Processing Algorithm: We have an independently developed brain - electrical signal processing algorithm that can effectively remove noise interference and accurately extract brain - electrical features related to cognitive

states. This algorithm has been verified through numerous experiments and has high accuracy and stability.

2. Neural Feedback Mechanism: Based on the analysis results of brain - electrical signals, a precise neural feedback mechanism is established. The system can adjust the training content and difficulty in a timely manner according to children's real - time brain - electrical states, achieving truly personalized training.

(II) Eye - Tracking Technology

1. Attention Analysis Model: Through in - depth analysis of eye - movement data, an attention analysis model is constructed. This model can accurately evaluate the attention distribution of children during the training process, providing an important basis for the formulation of personalized training plans.

(III) Multi - Sensory Collaborative Training Strategy

1. Cross - Modal Fusion Technology: Cross - modal fusion technology is used to organically integrate multiple sensory information such as vision, hearing, and touch. Through carefully designed training tasks, multiple sensory channels work together to promote neural remodeling and improve children's information processing ability and cognitive functions.

2. Personalized Sensory Stimulation Adjustment: According to children's individual sensory preferences and cognitive characteristics, the system can automatically adjust the intensity, frequency, and combination of multi - sensory stimuli to achieve a personalized sensory stimulation experience and improve the training effect.

Part 5: Team Introduction

(I) Core Members

1. Project Leader - Yin Zicheng: A sophomore undergraduate student majoring in Music Engineering. He/She has served as the person - in - charge and keynote speaker of many large - scale events, participated in the theme competition of the Pasteur Institute and won awards, carried out several scientific research projects, and won a total of 11 district - level or above awards. He/She is proficient in using audio software and hardware and the C programming language, with rich experience in software development and UI design.

2. Design and Development - Zhang Jiangnan: Graduated from Fudan University with a bachelor's degree and is currently a postgraduate student majoring in Music Technology at Shanghai Conservatory of Music. He/She is proficient in using various audio

software and hardware, currently works at Tianqiao Institute of Brain Science, and has participated in the development of the closed - loop binaural beat and sound wave research platform project.

3. Promotion and Publicity - Tao Yitian: A sophomore undergraduate student majoring in Music Engineering. He/She is proficient in using various audio software, has conducted research on music - based medical treatment, and participated in the publication of relevant papers. He/She has also participated in the organization and publicity of the Shanghai Musical Instrument Exhibition and is involved in front - end development material collection and later publicity work in this project.

4. External Liaison - Wu Qiwen: A freshman undergraduate student majoring in Music Engineering. He/She has participated in the AI music evaluation project of ByteDance and the production of relevant programs for the healing project of the Eye, Ear, Nose and Throat Hospital Affiliated to Fudan University. He/She has served as a brand ambassador for the Shanghai Musical Instrument Exhibition and a commentator in the MIDI area, with rich external experience.

(II) Team Advantages

The team members come from different fields and possess interdisciplinary knowledge and skills. At the same time, they have an innovative spirit and are passionate about the field of rehabilitation training for children with cognitive disorders, committed to providing the best rehabilitation solutions for these children.

Part 6: Profit Model

(I) Product Sales

The rehabilitation training answering system is sold at a reasonable price to medical institutions and families of children with cognitive disorders that have the conditions. For medical institutions, customized system services are provided. The system is personalized configured according to their scale and requirements, and corresponding fees are charged. For families of children with disorders, different versions of home - use products are launched to meet different levels of consumer demand.

(II) System Maintenance and Upgrade Services

System maintenance and upgrade services are provided to customers, and certain service fees are charged. The system is regularly maintained to ensure its stable operation. According to

technological development and user feedback, upgraded versions of the system are launched in a timely manner, adding new functions and training content to enhance the user experience and continuously create value for customers.

(III) Data Services

On the premise of being legal and compliant, in - depth analysis and mining are carried out on a large amount of training data collected by the system. Data services are provided to medical institutions, scientific research institutions, etc., such as providing data analysis reports related to cognitive disorder rehabilitation and assisting in scientific research projects, from which revenue can be obtained.

Part 7: Project Progress

(I) The First Stage

We completed the preliminary research on the rehabilitation training market for children with cognitive disorders, understood the limitations of existing rehabilitation training methods and market demands. We also communicated with some medical institutions and scientific research institutions and obtained valuable opinions and suggestions. At the same time, the project team was

established, and the responsibilities and divisions of each member were clarified.

(II) The Second Stage

We refined the system requirements and designed the technical solution. We maintained close cooperation with hospitals and Shanghai Tianqiao Institute of Brain Science, obtaining professional data resources and technical support. Meanwhile, we carried out the selection and procurement of hardware equipment to prepare for the system development.

(III) The Third Stage

We developed the 1.0 version of the answering system and conducted preliminary tests at Shanghai Children's Hospital. We optimized the algorithm model to improve the accuracy and efficiency of training. We planned to conduct small - scale clinical trials, collect preliminary feedback from children and medical staff, and improve the system according to the feedback results.

(IV) The Fourth Stage

We will comprehensively expand the scale of clinical trials to further verify the effectiveness and safety of the system. Simultaneously,

we will prepare academic papers to summarize research results and technological innovation points. We will complete the final debugging and optimization of the system to fully prepare for product promotion. At the same time, we will actively seek cooperation with more medical institutions and partners to expand market channels.